

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE BEST DIRECTION

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The Best Direction

Suppose we need to map our data $x \in \mathbb{R}^d$ into just **one** dimension:

$$x \mapsto u \cdot x \quad \text{for some unit direction } u \in \mathbb{R}^d$$

What is the direction u of maximum variance?

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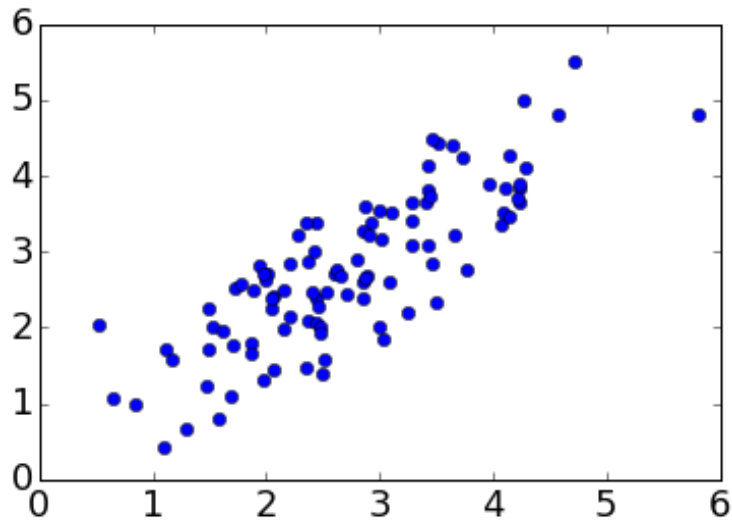
What is the direction u of maximum variance?

Useful fact 1:

- Let Σ be the $d \times d$ covariance matrix of X .
- The variance of X in direction u (the variance of $X \cdot u$) is:

$$u^T \Sigma u.$$

Best Direction Example



Here covariance matrix $\Sigma = \begin{pmatrix} 1 & 0.85 \\ 0.85 & 1 \end{pmatrix}$

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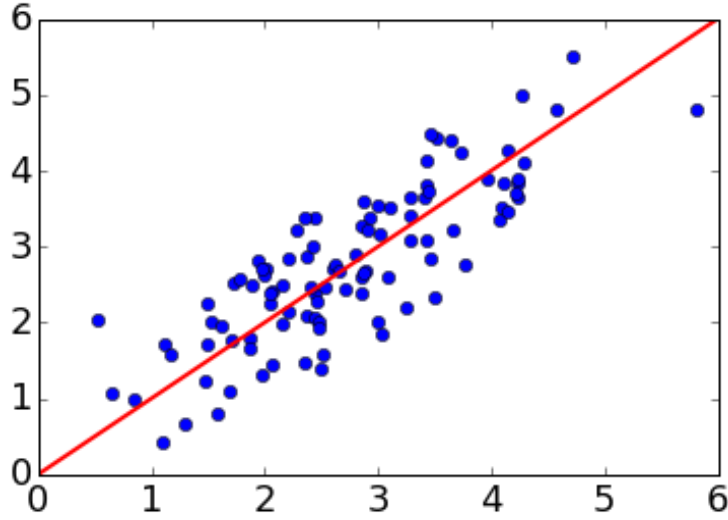
Useful fact 1:

- Let Σ be the $d \times d$ covariance matrix of X .
- The variance of X in direction u is given by $u^T \Sigma u$.

Useful fact 2:

- $u^T \Sigma u$ is maximized by setting u to the first **eigenvector** of Σ .
- The maximum value is the corresponding **eigenvalue**.

Best Direction Example



Direction: **first eigenvector** of the 2×2 covariance matrix of the data.

Projection onto this direction: the top **principal component** of the data